

Society App – Master Document

Society / building maintenance management · Resident + Staff app and Web · Complete A to Z blueprint (working name: SocietyHub)

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1. Overview & goals

- **Product:** One platform to run a housing society: bills, complaints, visitors, notices, bookings, accounts and records sharing.
- **Platforms:** Mobile app (Android and iOS), Website (public site and resident login) and Admin web panel. All use one backend.
- **Users:** Residents (owners and tenants), committee, manager, accountant, security guards and the society admin.
- **Key idea:** Role-based access. Everyone who registers is a Resident by default, and the Admin changes roles from the HR module.

2. Roles & registration rule

Role	Purpose
Resident (default)	Owner or tenant. Pays bills, raises issues, invites visitors, books amenities, shares own records.
Admin	Full control. Assigns or removes roles in the HR module, sets rules, sees everything.
Manager	Handles complaints, bookings, notices and staff.
Accountant	Bills, collections, expenses, ledger and reports.
Committee	Polls, meetings, read-only view of accounts and complaints.
Guard	Scans gate pass, logs visitors, checks staff and vehicles.

Registration rule

- Register with name, mobile, society and flat, then OTP. The user becomes **Resident** immediately.

There is no approval step.

- Admin opens the **HR module** at any time to assign a role, change a role, remove a role (back to Resident) or remove the user from the society.
- Every role change is saved in the audit log (who, whom, when).

3. Module list

Module	What it does
Auth & Profile	OTP login, register, profile, family members, vehicles
HR & Roles	Users list, assign or remove roles, permissions, audit log
Accounts & Billing	Generate bills, pay online, collections, expenses, ledger, receipts
Rent & Records	Tenant rent, receipts, charges, payment history, statements
Sharing	Share receipts, issues and statements by PDF, link or invited person (landlord or family)
Complaints	Raise, assign, track, resolve, rate
Visitors & Gate	Invite, QR gate pass, walk-in approval, in/out log, staff and vehicle check
Notices & Polls	Announcements, polls, results, meetings and RSVP
Amenities	Slots, booking, fees, approvals
Directory & Documents	Society directory, important contacts, document library
Notifications	Push, SMS or WhatsApp, in-app list
Reports	Collection, defaulters, income and expense, balance sheet

4. Screen inventory (all screens)

Total 56 screens. IDs are used across wireframe pages and can be used as ticket numbers.

A. Common (all roles)

ID	Screen	Notes
A1	Login with OTP	Mobile number, OTP verify
A2	Register	Name, mobile, society, flat, owner or tenant
A3	Home dashboard	Menu tiles change by role
A4	Notifications	Bills, visitors, complaints, polls

B. Resident modules

ID	Screen	Notes
B1	My accounts / dues	Bill breakup, pay via UPI, card, net banking
B2	Complaints list and raise new	Category, photo, status filter
B3	My visitors and gate pass	Invite, QR pass, share
B4	Visitor log	In/out history, allow or deny
B5	Notice board	Notices, RSVP
B6	Polls (vote)	Options, vote, result
B7	Amenity booking	Amenity, date, slot, fee
B8	My bookings	Cancel or reschedule
B9	Society directory	Search, call
B10	Important contacts	Security, plumber, electrician, emergency
B11	Document library	Bylaws, minutes, audit reports
B12	Family members	Add, edit
B13	My vehicles	Add, parking slot
B14	Help and support	Contact society office

C. Rent, records & sharing (Resident)

ID	Screen	Notes
C1	My records hub	Rent, charges, payments, issues, summary
C2	Rent	Monthly rent, pay, mark paid, agreement
C3	Rent receipt	Generate, share, PDF
C4	Charges breakup	Maintenance, water, parking, late fee
C5	Payment history	Filter, tap to open receipt
C6	Issue detail and share	Timeline, photos, comment
C7	Export statement	Date range, what to include, PDF or Excel
C8	Share sheet	WhatsApp, email, link, PDF, expiry, OTP
C9	Shared access	People and links, revoke

C10	Invite person	Landlord or family, choose what they see
C11	Recipient (landlord) view	Read-only tenant rent status

D. Admin

ID	Screen	Notes
D1	Admin dashboard	Collections, pending, complaints, visitors
D2	HR: users list	Search, filter by role
D3	User details	Change role, remove role, remove user, role history
D4	Assign role	Choose role and save
D5	Roles and permissions	Edit what each role can do
D6	Society settings	Profile, flats, rates, late fee, amenities, gateway
D7	Audit log	All role and data changes

E. Manager

ID	Screen	Notes
E1	Manager home	New complaints, pending bookings
E2	Complaint queue	Filter, priority
E3	Assign and update complaint	Staff, due date, resolve
E4	Booking approvals	Approve or reject, slot calendar
E5	Create notice	Audience, attachment, push

F. Accountant

ID	Screen	Notes
F1	Accounts home	Month collection, defaulters
F2	Generate bills	Month, flats, heads, due date
F3	Collections	Paid, pending, overdue, cash entry, reminder
F4	Expenses	Add expense with bill photo
F5	Reports	Download PDF or Excel

G. Committee

ID	Screen	Notes
G1	Committee home	Tiles, next meeting
G2	Create poll	Question, options, audience, end date
G3	Poll results	Percent, votes, export
G4	Meetings	Schedule, RSVP, minutes

H. Security guard

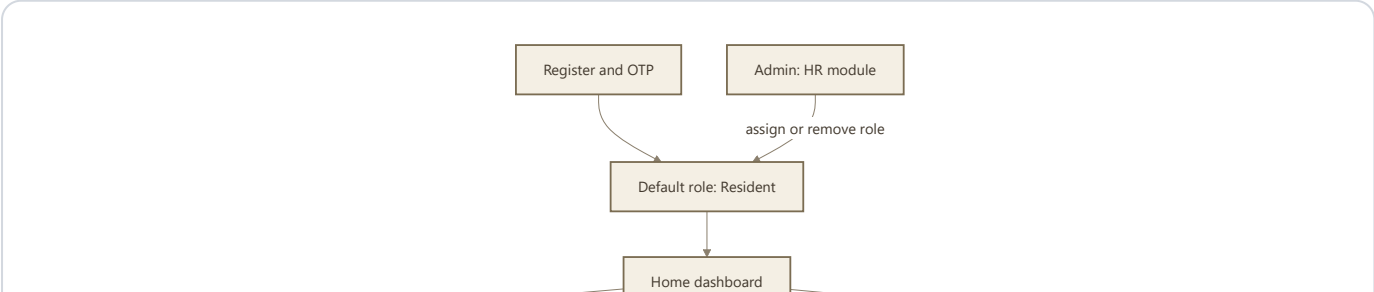
ID	Screen	Notes
H1	Guard home	QR scan, inside count
H2	Walk-in entry	Name, purpose, flat, photo
H3	Approval status	Resident approve or deny
H4	Visitor in/out log	Mark exit
H5	Staff and vehicle check	Valid or not, emergency alert

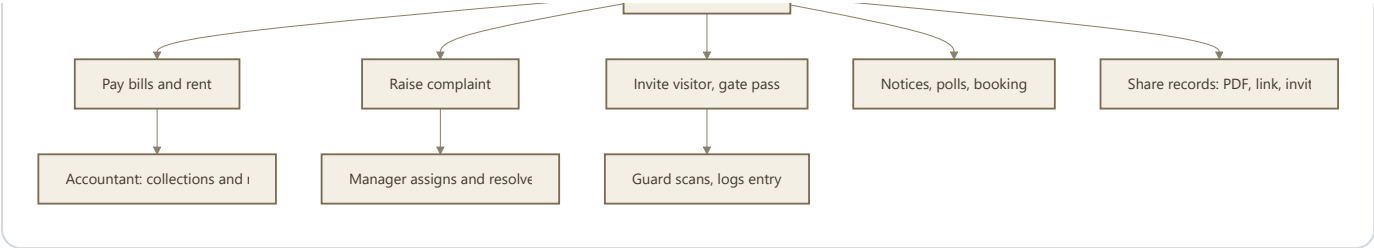
W. Website and web panels

ID	Screen	Notes
W1	Landing page	Hero, features, pricing, demo form
W2	Resident web login and dashboard	Same modules as app
W3	Admin web panel	Sidebar, KPIs, tables, HR module

5. Process flows

Master flow





- **Bill payment:** Accountant creates bills, resident is notified, pays online, receipt and ledger update, reminder and late fee after due date.
- **Complaint:** Raised (Open), manager assigns (In progress), staff fixes, manager marks Resolved, resident confirms and rates (Closed) or reopens.
- **Visitor:** Resident invites, QR pass, guard scans, entry logged and resident notified, exit logged. Without a pass the guard requests approval.
- **Sharing:** Pick a record, share as PDF, secure link (expiry, view only, OTP) or invite a person, recipient views read-only, resident can revoke.
- **Booking:** Pick slot, pay fee or wait for approval, confirmed, reminder, cancel or reschedule.

6. Permission matrix

Action	Resident	Guard	Accountant	Committee	Manager	Admin
HR: assign or remove roles and users	–	–	–	–	–	✓
Pay bills, view own dues	✓	–	✓	✓	✓	✓
Create bills, accounts, reports	–	–	✓	View	View	✓
Raise complaint	✓	✓	✓	✓	✓	✓
Assign or close complaints	–	–	–	–	✓	✓
Invite visitor	✓	–	✓	✓	✓	✓
Scan pass, visitor in/out	–	✓	–	–	✓	✓
Post notices	–	–	–	✓	✓	✓
Create polls (all can vote)	Vote	Vote	Vote	✓	✓	✓
Book amenity	✓	–	✓	✓	✓	✓
Approve bookings	–	–	–	–	✓	✓
Share own rent, payments, issues	✓	✓	✓	✓	✓	✓

Permissions are stored in the database, so roles can be edited without code changes. The backend always checks them.

7. Data model

```
users(id, name, phone, email, status)           -- ACTIVE / REMOVED
societies(id, name, address)
flats(id, society_id, tower, number)
roles(id, society_id, name)                     -- Admin, Manager, Guard...
permissions(id, code)                           -- 'complaint.assign'
role_permissions(role_id, permission_id)
user_roles(user_id, society_id, flat_id, role_id) -- default: Resident
audit_log(id, actor_id, action, target_id, at)
bills(id, flat_id, month, heads_json, amount, due_date, status)
payments(id, bill_id|rent_id, payer_id, amount, mode, receipt_no, paid_at)
rent_agreements(id, flat_id, tenant_id, landlord_name, monthly_rent, start, end)
rent_payments(id, agreement_id, month, amount, mode, status)
complaints(id, flat_id, category, priority, status, assigned_to)
complaint_updates(id, complaint_id, note, photo, at)
visitors(id, flat_id, name, purpose, pass_qr, in_at, out_at, status)
notices(id, title, body, audience) polls(...) poll_votes(...)
amenities(id, name, fee) bookings(id, amenity_id, user_id, slot, status)
shares(id, owner_id, record_type, record_id, token, expires_at, access, revoked)
share_invites(id, owner_id, invitee_phone, relation, scopes_json, revoked)
documents(id, title, file_url) notifications(id, user_id, body, read)
```

8. Tech stack & APIs

- **App:** Flutter (Android and iOS). **Web and admin:** React or Next.js. **Backend:** Node.js (NestJS) with PostgreSQL.
- **Payments:** Razorpay (UPI, cards, net banking). **Notifications:** Firebase push plus SMS or WhatsApp OTP. **Files:** S3 storage.
- **API groups:** /auth /users /roles /bills /payments /rent /complaints /visitors /notices /polls /amenities /shares /reports
- **Security:** OTP login with JWT, permission check on every API, audit log, secure share links with expiry and revoke, and private data shared only by the owner.

9. Build phases

Phase	Scope
Phase 1 (MVP)	Auth, register, HR roles and permissions, home, notices, complaints, bills and online payment, profile
Phase 2	Visitors and gate pass, guard app, amenities, polls, notifications, directory and contacts
Phase 3	Rent and records, sharing (PDF, link, invite), statements, expenses, reports, documents, meetings

Phase 4

Website and admin web panel polish, analytics, multi-society super admin, real estate listings (optional)

10. Visual wireframe pages

Open these for the screen drawings that match the IDs above

Resident app screens

A1–A5, B1–B14

Rent, records and sharing

C1–C11

Admin, Manager, Accountant, Committee, Guard

D1–H5

Website and admin web

W1–W3

Process flows

Registration, HR, module flows